Practice M6: FHS, Disks, Filesystems, and Archives (Additional)

* NOTE: Most of the exercises included in this practice guide **are not suitable** for execution in **WSL** or **Docker** environments. It is recommended that a virtual infrastructure to be used instead.

For the purpose of the current practice, we will need one virtual machine powered on, but for different parts, we will use different machine based on different distribution.

All commands that we will use until the end of this practice will be accompanied by the appropriate prompt. This way it will be easier for us to know which user, in which folder, and on which machine is executing the command.

Every part will state clearly which distribution is used.

Btrfs and openSUSE

Let's check where we start from:

```
lsauser@opensuse:~> lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda      8:0    0    8G  0 disk
├─sda1   8:1    0    8M  0 part
└─sda2   8:2    0    8G  0 part /
sr0     11:0    1 1024M  0 rom
lsauser@opensuse:~>
```

Now, let's turn off the **VM** and add four additional hard disks:

- First with size of **1GB**
- Second with size of **2GB**
- Third with size of **500MB**
- Fourth with size of **500MB**

Then power on the **VM** again and check the situation:

```
lsauser@opensuse:~> lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda      8:0    0    8G  0 disk
├─sda1   8:1    0    8M  0 part
└─sda2   8:2    0    8G  0 part /
sdb      8:16   0    1G  0 disk
sdc      8:32   0    2G  0 disk
sdd      8:48   0  512M  0 disk
sde      8:64   0  512M  0 disk
sr0     11:0    1 1024M  0 rom
lsauser@opensuse:~>
```

Our new disks are listed and reachable via **sdb**, **sdc**, **sdd**, and **sde** identifiers.

Now, we can create a filesystem that in a similar way to **LVM** spans over multiple drives with different sizes. Please note that if one of the drives fails, you will lose all information.

Let's use the **sdb** and **sdc** drives:

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```
lsauser@opensuse:~> sudo mkfs.btrfs -d single -f -L VOLUME-BTRFS /dev/sd[bc]
btrfs-progs v4.15
See http://btrfs.wiki.kernel.org for more information.

Label:              VOLUME-BTRFS
UUID:               fd667810-f59d-4d0f-bf2d-b8228efb2315
Node size:          16384
Sector size:        4096
Filesystem size:    3.00GiB
Block group profiles:
  Data:             single            8.00MiB
  Metadata:         RAID1           153.56MiB
  System:           RAID1             8.00MiB
SSD detected:       no
Incompat features:  extref, skinny-metadata
Number of devices:  2
Devices:
   ID        SIZE  PATH
    1     1.00GiB  /dev/sdb
    2     2.00GiB  /dev/sdc

lsauser@opensuse:~>
```

As we can see, metadata and system data are stored in **RAID1** mode. This means that those will be mirrored across all drives. We can force them (system + metadata) to have only one copy and follow the data distribution (mode) by adding a **-M** switch.

Similarly, we can create any of the supported **RAID** levels. For example, we can create a striped **RAID** or **RAID0** with:

```
lsauser@opensuse:~> sudo mkfs.btrfs -d raid0 -f -L RAID0-BTRFS /dev/sd[de]
btrfs-progs v4.15
See http://btrfs.wiki.kernel.org for more information.

Label:              RAID0-BTRFS
UUID:               808842cd-afc2-42fb-9184-c5754ba7d3ea
Node size:          16384
Sector size:        4096
Filesystem size:    1.00GiB
Block group profiles:
  Data:             RAID0           128.00MiB
  Metadata:         RAID1            51.19MiB
  System:           RAID1             8.00MiB
SSD detected:       no
Incompat features:  extref, skinny-metadata
Number of devices:  2
Devices:
   ID        SIZE  PATH
```

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```
    1   512.00MiB  /dev/sdd
    2   512.00MiB  /dev/sde

lsauser@opensuse:~>
```

Should we want to change the mode data is stored, we must change the value after the **-d** switch to **raid1**, **raid5** or something else. In the same way, we can force the system and metadata to follow the data mode.

Now the mounting is the same as with other filesystems:

```
lsauser@opensuse:~> sudo mkdir -p /disks/btrfs-{volume,raid}
lsauser@opensuse:~> sudo mount /dev/sdb /disks/btrfs-volume
lsauser@opensuse:~> sudo mount /dev/sdd /disks/btrfs-raid
```

We can use any drive member of the appropriate group for mounting.

If we execute the **lsblk** command, we will get:

```
lsauser@opensuse:~> lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda      8:0    0    8G  0 disk
├─sda1   8:1    0    8M  0 part
└─sda2   8:2    0    8G  0 part /
sdb      8:16   0    1G  0 disk /disks/btrfs-volume
sdc      8:32   0    2G  0 disk
sdd      8:48   0  512M  0 disk /disks/btrfs-raid
sde      8:64   0  512M  0 disk
sr0     11:0    1 1024M  0 rom
lsauser@opensuse:~>
```

One way to ask for free space is to use the well-known command **df**:

```
lsauser@opensuse:~> df -h /disks/btrfs-*
Файлова с-ма   Размер Заети Свобод Изп% Монтирана на
/dev/sdd         1,0G   17M   904M   2% /disks/btrfs-raid
/dev/sdb         3,0G   17M   2,7G   1% /disks/btrfs-volume
lsauser@opensuse:~>
```

An alternative way is to use the **btrfs** command. For example:

```
lsauser@opensuse:~> sudo btrfs filesystem show
Label: 'VOLUME-BTRFS'  uuid: fd667810-f59d-4d0f-bf2d-b8228efb2315
        Total devices 2 FS bytes used 192.00KiB
        devid    1 size 1.00GiB used 169.56MiB path /dev/sdb
        devid    2 size 2.00GiB used 161.56MiB path /dev/sdc

Label: 'RAID0-BTRFS'  uuid: 808842cd-afc2-42fb-9184-c5754ba7d3ea
        Total devices 2 FS bytes used 128.00KiB
        devid    1 size 512.00MiB used 123.19MiB path /dev/sdd
        devid    2 size 512.00MiB used 123.19MiB path /dev/sde
```

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```
lsauser@opensuse:~>
```

If we want to automount both filesystems by adding them to the **/etc/fstab**, we can use either the **UUID** value or any of the member devices, as we did earlier.

We can ask for detailed information for one of the filesystems with:

```
lsauser@opensuse:~> sudo btrfs filesystem usage /disks/btrfs-raid
Overall:
    Device size:                   1.00GiB
    Device allocated:            246.38MiB
    Device unallocated:          777.62MiB
    Device missing:                  0.00B
    Used:                        256.00KiB
    Free (estimated):            905.62MiB      (min: 516.81MiB)
    Data ratio:                       1.00
    Metadata ratio:                   2.00
    Global reserve:               16.00MiB      (used: 0.00B)

Data,RAID0: Size:128.00MiB, Used:0.00B
   /dev/sdd       64.00MiB
   /dev/sde       64.00MiB

Metadata,RAID1: Size:51.19MiB, Used:112.00KiB
   /dev/sdd       51.19MiB
   /dev/sde       51.19MiB

System,RAID1: Size:8.00MiB, Used:16.00KiB
   /dev/sdd        8.00MiB
   /dev/sde        8.00MiB

Unallocated:
   /dev/sdd      388.81MiB
   /dev/sde      388.81MiB
lsauser@opensuse:~>
```

Or use a sub-command similar to **df**:

```
lsauser@opensuse:~> sudo btrfs filesystem df /disks/btrfs-raid
Data, RAID0: total=128.00MiB, used=0.00B
System, RAID1: total=8.00MiB, used=16.00KiB
Metadata, RAID1: total=51.19MiB, used=112.00KiB
GlobalReserve, single: total=16.00MiB, used=0.00B
lsauser@opensuse:~>
```

Okay, but the filesystem is empty. We can create a file full of zeroes with:

```
lsauser@opensuse:~> sudo dd if=/dev/zero of=/disks/btrfs-raid/file.dat bs=1M count=200
200+0 прочетени блока
```

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```
200+0 записани блока
209715200 bytes (210 MB, 200 MiB) copied, 0,303935 s, 690 MB/s
lsauser@opensuse:~>
```

Now, we can execute first:

```
lsauser@opensuse:~> ls -alh /disks/btrfs-raid/
общо 201M
drwxr-xr-x 1 root root   16 13 мар 16,23 .
drwxr-xr-x 4 root root 4,0K 13 мар 16,13 ..
-rw-r--r-- 1 root root 200M 13 мар 16,23 file.dat
lsauser@opensuse:~>
```

Then we can try with the regular **df** command:

```
lsauser@opensuse:~> df -h /disks/btrfs-raid/
Файлова с-ма   Размер Заети Свобод Изп% Монтирана на
/dev/sdd         1,0G  217M   704M  24% /disks/btrfs-raid
lsauser@opensuse:~>
```

It can take some time to reflect the changes. We can force it with **sync**. More or less similar out we can get from the:

```
lsauser@opensuse:~> sudo btrfs filesystem df /disks/btrfs-raid
Data, RAID0: total=256.00MiB, used=200.12MiB
System, RAID1: total=8.00MiB, used=16.00KiB
Metadata, RAID1: total=51.19MiB, used=320.00KiB
GlobalReserve, single: total=16.00MiB, used=0.00B
lsauser@opensuse:~>
```

And if we want really detailed information, we can use:

```
lsauser@opensuse:~> sudo btrfs filesystem usage /disks/btrfs-raid
Overall:
    Device size:                   1.00GiB
    Device allocated:            374.38MiB
    Device unallocated:          649.62MiB
    Device missing:                  0.00B
    Used:                        200.78MiB
    Free (estimated):            705.50MiB      (min: 380.69MiB)
    Data ratio:                       1.00
    Metadata ratio:                   2.00
    Global reserve:               16.00MiB      (used: 0.00B)

Data,RAID0: Size:256.00MiB, Used:200.12MiB
   /dev/sdd      128.00MiB
   /dev/sde      128.00MiB

Metadata,RAID1: Size:51.19MiB, Used:320.00KiB
   /dev/sdd       51.19MiB
   /dev/sde       51.19MiB
```

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```
System,RAID1: Size:8.00MiB, Used:16.00KiB
   /dev/sdd        8.00MiB
   /dev/sde        8.00MiB

Unallocated:
   /dev/sdd      324.81MiB
   /dev/sde      324.81MiB
lsauser@opensuse:~>
```

RAID with MDRaid

For this part we will use an **AlmaLinux VM**. Let's check where we start from:

```
[lsauser@almalinux ~]$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda                   8:0    0    8G  0 disk
├─sda1                8:1    0    1G  0 part /boot
└─sda2                8:2    0    7G  0 part
  ├─almalinux-root 253:0    0  6,2G  0 lvm  /
  └─almalinux-swap 253:1    0  820M  0 lvm  [SWAP]
sr0                  11:0    1 1024M  0 rom
[lsauser@almalinux ~]$
```

Now, let's turn off the **VM** and add four additional hard disks each with **1GB** of size.

Then power on the **VM** again and check the situation:

```
[lsauser@almalinux ~]$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda                   8:0    0    8G  0 disk
├─sda1                8:1    0    1G  0 part /boot
└─sda2                8:2    0    7G  0 part
  ├─almalinux-root 253:0    0  6,2G  0 lvm  /
  └─almalinux-swap 253:1    0  820M  0 lvm  [SWAP]
sdb                   8:16   0    1G  0 disk
sdc                   8:32   0    1G  0 disk
sdd                   8:48   0    1G  0 disk
sde                   8:64   0    1G  0 disk
sr0                  11:0    1 1024M  0 rom
[lsauser@almalinux ~]$
```

We should check if **mdadm** is installed, and if not – install it with:

```
[lsauser@almalinux ~]$ sudo dnf install -y mdadm
...
```

Use the package manager for your distribution.

Before we can continue with **RAID** creation, we must prepare the disks. For this purpose, we will create one primary partition on each disk with type **fd** (Linux raid autodetect). The procedure for the first spare drive (**/dev/sdb**) is:

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```
[lsauser@almalinux ~]$ sudo fdisk /dev/sdb
...
Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-2097151, default 2048):
Using default value 2048
Last sector, +sectors or +size{K,M,G} (2048-2097151, default 2097151):
Using default value 2097151
Partition 1 of type Linux and of size 1023 MiB is set

Command (m for help): t
Selected partition 1
Hex code (type L to list all codes): fd
Changed type of partition 'Linux' to 'Linux raid autodetect'

Command (m for help): p

Disk /dev/sdb: 1073 MB, 1073741824 bytes, 2097152 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x08e5f00b

   Device Boot      Start         End      Blocks   Id  System
/dev/sdb1            2048     2097151     1047552   fd  Linux raid autodetect

Command (m for help): w
The partition table has been altered!

Calling ioctl() to re-read partition table.
Syncing disks.
[lsauser@almalinux ~]$
```

We must repeat this for the other three spare drives (**/dev/sdc**, **/dev/sdd** and **/dev/sde**).

An alternative approach is to follow the scripted approach and use the **parted** utility. In fact, we will execute four similar commands, which will contain all the arguments needed to fulfill the task:

```
[lsauser@almalinux ~]$ sudo parted -s /dev/sdb -- mklabel msdos mkpart primary 1m -0m set 1 raid on
[lsauser@almalinux ~]$ sudo parted -s /dev/sdc -- mklabel msdos mkpart primary 1m -0m set 1 raid on
[lsauser@almalinux ~]$ sudo parted -s /dev/sdd -- mklabel msdos mkpart primary 1m -0m set 1 raid on
[lsauser@almalinux ~]$ sudo parted -s /dev/sde -- mklabel msdos mkpart primary 1m -0m set 1 raid on
```

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Either way – interactive approach with **fdisk**, or scripted approach with **parted**, at the end we should have four spare drives each with single partition. The **lsblk** output should now be similar to this:

```
[lsauser@almalinux ~]$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda                   8:0    0    8G  0 disk
├─sda1                8:1    0    1G  0 part /boot
└─sda2                8:2    0    7G  0 part
  ├─almalinux-root 253:0    0  6,2G  0 lvm  /
  └─almalinux-swap 253:1    0  820M  0 lvm  [SWAP]
sdb                   8:16   0    1G  0 disk
└─sdb1                8:17   0 1023M  0 part
sdc                   8:32   0    1G  0 disk
└─sdc1                8:33   0 1023M  0 part
sdd                   8:48   0    1G  0 disk
└─sdd1                8:49   0 1023M  0 part
sde                   8:64   0    1G  0 disk
└─sde1                8:65   0 1023M  0 part
sr0                  11:0    1 1024M  0 rom
[lsauser@almalinux ~]$
```

We are ready to proceed with **RAID** creation. First, we will create one **RAID0** array:

```
[lsauser@almalinux ~]$ sudo mdadm --create /dev/md0 --level=0 --raid-devices=2 /dev/sdb1 /dev/sdc1
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
```

We can check **RAID** status either with this:

```
[lsauser@almalinux ~]$ cat /proc/mdstat
Personalities : [raid0]
md0 : active raid0 sdc1[1] sdb1[0]
      2091008 blocks super 1.2 512k chunks

unused devices: <none>
```

Or we can ask for more detailed information with this:

```
[lsauser@almalinux ~]$ sudo mdadm -D /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Thu Mar 14 12:16:55 2019
        Raid Level : raid0
        Array Size : 2091008 (2042.00 MiB 2141.19 MB)
      Raid Devices : 2
     Total Devices : 2
       Persistence : Superblock is persistent

       Update Time : Thu Mar 14 12:16:55 2019
```

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```
             State : clean
    Active Devices : 2
   Working Devices : 2
    Failed Devices : 0
     Spare Devices : 0

        Chunk Size : 512K

Consistency Policy : none

              Name : almalinux:0  (local to host almalinux)
              UUID : c2321fb6:799c84a3:1c3839dc:4b5ba067
            Events : 0

    Number   Major   Minor   RaidDevice State
       0       8       17        0      active sync   /dev/sdb1
       1       8       33        1      active sync   /dev/sdc1
[lsauser@almalinux ~]$
```

In a similar fashion we can create **RAID1** array out of the other two spare disks:

```
[lsauser@almalinux ~]$ sudo mdadm --create /dev/md1 --level=1 --raid-devices=2 /dev/sdd1 /dev/sde1
mdadm: Note: this array has metadata at the start and
    may not be suitable as a boot device.  If you plan to
    store '/boot' on this device please ensure that
    your boot-loader understands md/v1.x metadata, or use
    --metadata=0.90
Continue creating array? yes
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md1 started.
[lsauser@almalinux ~]$ cat /proc/mdstat
Personalities : [raid0] [raid1]
md1 : active raid1 sde1[1] sdd1[0]
      1046528 blocks super 1.2 [2/2] [UU]

md0 : active raid0 sdc1[1] sdb1[0]
      2091008 blocks super 1.2 512k chunks

unused devices: <none>
[lsauser@almalinux ~]$ sudo mdadm -D /dev/md1
/dev/md1:
           Version : 1.2
     Creation Time : Thu Mar 14 12:24:13 2019
        Raid Level : raid1
        Array Size : 1046528 (1022.00 MiB 1071.64 MB)
     Used Dev Size : 1046528 (1022.00 MiB 1071.64 MB)
```

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```
      Raid Devices : 2
     Total Devices : 2
       Persistence : Superblock is persistent

       Update Time : Thu Mar 14 12:24:16 2019
             State : clean
    Active Devices : 2
   Working Devices : 2
    Failed Devices : 0
     Spare Devices : 0

Consistency Policy : resync

              Name : almalinux:1  (local to host almalinux)
              UUID : c0c75f3a:75857e0c:e79bd352:2e25e062
            Events : 17

    Number   Major   Minor   RaidDevice State
       0       8       49        0      active sync   /dev/sdd1
       1       8       65        1      active sync   /dev/sde1
[lsauser@almalinux ~]$
```

Now, we must take one more step to ensure that both RAID arrays will be reassembled on boot (the file for **Debian/Ubuntu** is **/etc/mdadm/mdadm.conf**):

```
[lsauser@almalinux ~]$ sudo mdadm --detail --scan | sudo tee -a /etc/mdadm.conf
ARRAY /dev/md/0 metadata=1.2 name=almalinux:0 UUID=c2321fb6:799c84a3:1c3839dc:4b5ba067
ARRAY /dev/md/1 metadata=1.2 name=almalinux:1 UUID=c0c75f3a:75857e0c:e79bd352:2e25e062
[lsauser@almalinux ~]$
```

It is time to create filesystems for each **RAID** array. Here we will use ext4:

```
[lsauser@almalinux ~]$ sudo mkfs.ext4 /dev/md0
...
[lsauser@almalinux ~]$ sudo mkfs.ext4 /dev/md1
...
[lsauser@almalinux ~]$
```

Then, we must prepare the mount points:

```
[lsauser@almalinux ~]$ sudo mkdir -p /disks/raid{0,1}
```

The final step is to add records in the **/etc/fstab** file. This can be done either by opening the file in text editor, or by appending information in it with:

```
[lsauser@almalinux ~]$ echo '/dev/md0 /disks/raid0 ext4 defaults 0 0' | sudo tee -a /etc/fstab
/dev/md0 /disks/raid0 ext4 defaults 0 0
[lsauser@almalinux ~]$ echo '/dev/md1 /disks/raid1 ext4 defaults 0 0' | sudo tee -a /etc/fstab
/dev/md1 /disks/raid1 ext4 defaults 0 0
[lsauser@almalinux ~]$
```

Now, we can reboot, and once the system is up and running again, we can check how both arrays are seen:

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```
[lsauser@almalinux ~]$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINT
sda                   8:0    0    8G  0 disk
├─sda1                8:1    0    1G  0 part  /boot
└─sda2                8:2    0    7G  0 part
  ├─almalinux-root 253:0    0  6,2G  0 lvm   /
  └─almalinux-swap 253:1    0  820M  0 lvm   [SWAP]
sdb                   8:16   0    1G  0 disk
└─sdb1                8:17   0 1023M  0 part
  └─md0               9:0    0    2G  0 raid0 /disks/raid0
sdc                   8:32   0    1G  0 disk
└─sdc1                8:33   0 1023M  0 part
  └─md0               9:0    0    2G  0 raid0 /disks/raid0
sdd                   8:48   0    1G  0 disk
└─sdd1                8:49   0 1023M  0 part
  └─md1               9:1    0 1022M  0 raid1 /disks/raid1
sde                   8:64   0    1G  0 disk
└─sde1                8:65   0 1023M  0 part
  └─md1               9:1    0 1022M  0 raid1 /disks/raid1
sr0                  11:0    1 1024M  0 rom
[lsauser@almalinux ~]$ df -h
Filesystem                  Size  Used Avail Use% Mounted on
/dev/mapper/almalinux-root  6,2G  1,2G  5,1G  19% /
devtmpfs                    484M     0  484M   0% /dev
tmpfs                       496M     0  496M   0% /dev/shm
tmpfs                       496M  6,8M  489M   2% /run
tmpfs                       496M     0  496M   0% /sys/fs/cgroup
/dev/sda1                  1014M  132M  883M  13% /boot
/dev/md0                    2,0G  6,0M  1,9G   1% /disks/raid0
/dev/md1                    990M  2,6M  921M   1% /disks/raid1
tmpfs                       100M     0  100M   0% /run/user/1000
[lsauser@almalinux ~]$
```

Using Physical USB Device

In this exercise we will take all necessary steps to use a physical **USB** flash drive formatted with **FAT32** in a **VM** running in **VirtualBox**. The steps are more or less the same in other virtualization solutions.

We can continue with the machine from the previous exercise or create/import a new one. It could be running any distribution, but next steps are executed on **AlmaLinux**.

Our **VM** must be in a running state. Then we must execute the following preparation steps:

- Plug in the **USB** stick
- Choose the **USB** drive from the **Devices > USB > Kingston USB** menu. Your **USB** drive most probably will have different name. This way we are attaching the device directly to the **VM**

Now, in a terminal session, execute:

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```
[lsauser@almalinux ~]$ lsblk
NAME            MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINT
...
sdf               8:80   1  3,8G  0 disk
└─sdf1            8:81   1  3,8G  0 part
...
[lsauser@almalinux ~]$
```

The name and size of the device will be different in your case.

We can create a dedicated folder to mount the **USB** drive's filesystem or use existing one. Let's create one:

```
[lsauser@almalinux ~]$ sudo mkdir -p /media/usb
```

Now that we know the path to the device, we can mount it:

```
[lsauser@almalinux ~]$ sudo mount /dev/sdf1 /media/usb
```

If we check the available block devices:

```
[lsauser@almalinux ~]$ lsblk -f
NAME            FSTYPE            LABEL    UUID                                   MOUNTPOINT
...
sdf
└─sdf1          vfat              DT 4GB   FA4D-608E                              /media/usb
...
[lsauser@almalinux ~]$
```

Let's create a small text file on the mounted flash drive:

```
[lsauser@almalinux ~]$ echo 'Hello from Linux!' | sudo tee /media/usb/hello.txt
Hello from Linux!
[lsauser@almalinux ~]$ ls -al /media/usb/
общо 460
drwxr-xr-x. 3 root root   4096 14 мар 17,33 .
drwxr-xr-x. 3 root root     17 14 мар 17,28 ..
-rwxr-xr-x. 1 root root     18 14 мар 17,33 hello.txt
-rwxr-xr-x. 1 root root 457281 13 мар 16,37 M6-Practice-FHS-Disks-Filesystems-and-Archives.docx
drwxr-xr-x. 2 root root   4096  9 дек 15,14 System Volume Information
[lsauser@almalinux ~]$
```

Now, let's unmount it:

```
[lsauser@almalinux ~]$ sudo umount /media/usb
[lsauser@almalinux ~]$ ls -al /media/usb/
общо 0
drwxr-xr-x. 2 root root  6 14 мар 17,28 .
drwxr-xr-x. 3 root root 17 14 мар 17,28 ..
[lsauser@almalinux ~]$
```

This gives us a valid option for data exchange between any **VM** and the host.

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